Modeling Bridge Deterioration Using Case-based Reasoning

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Modeling Bridge Deterioration Using Case-based Reasoning

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Abstract: Current bridge deterioration models used in bridge management systems are not successful in capturing the effects of bridge condition history on future condition, in performing ''what if'' analyses for different maintenance scenarios, and in accounting for the interactive effects between deterioration mechanisms of different bridge components. Moreover, these models cannot be easily updated when new data is obtained. On the other hand, bridge management systems are updated on a regular basis and thus accumulate valuable knowledge about the performance of bridges over the years. The case-based reasoning (CBR) approach could reuse this knowledge efficiently to act as a deterioration model that would overcome some of the shortcomings of current models. A new CBR system, called case-based reasoning for modeling infrastructure deterioration, was developed and utilized in building a ''proof of concept'' application for modeling the deterioration of concrete bridge decks using data obtained from the Canadian Province of Quebec. The performance of the CBR model showed that CBR has great potentials in predicting the future condition of infrastructure facilities.

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Introduction

Preserving the condition of highway bridges has become a crucial and growing concern since the late 1980s. Bridge management systems (BMSs) have been developed to optimize maintenance, rehabilitation, and replacement (MR&R) decisions for bridge networks under financial constraints. The quality of these decisions depends, to a great extent, on the ability to predict the future condition of bridges. This is why AASHTO has prescribed a bridge-deterioration model as one of the minimum requirements of any BMS (AASHTO 1993). By definition, ''a facility deterioration model links a measure of facility condition to a vector of explanatory variables'' (Ben-Akiva and Gopinath 1995). A measure of facility condition is an assessment of the extent and severity of facility damages on a numeric scale. Explanatory variables, such as age, traffic, and weather, are defined as the factors that affect facility deterioration and can be observed or measured. Several models have been developed since the early 1970s (specifically for pavements) to assist decision-makers in predicting the future condition of a network of facilities and, consequently, optimizing the allocation of the scarce resources on MRR needs of the network. These models can be grouped into three categories that are not mutually exclusive: deterministic models, stochastic models, and artificial intelligence models. These categories are listed in Table 1 and discussed below.

Deterministic models describe the relationship between the factors affecting facility deterioration (e.g., bridge age) and the facility condition using a mathematical or a statistical formulation. These models calculate the predicted conditions deterministically by ignoring the random error in prediction. Although deterministic models are efficient for the analysis of networks with a large population, they have the following limitations: (1) they neglect the uncertainty due to the inherent stochasticity of infrastructure deterioration and the existence of unobserved explanatory variables (Jiang and Sinha 1989; Madanat et al. 1995); (2) they predict the average condition of a family of facilities regardless of the current condition and the condition history of individual facilities (Shahin et al. 1987; Jiang and Sinha 1989); (3) they estimate facility deterioration for the ''no maintenance'' strategy only because of the difficulty of estimating the impacts of various maintenance strategies (Sanders and Zhang 1994); (4) they disregard the interaction between the deterioration mechanisms of different facility components such as between the bridge deck and the deck joints (Sianipar and Adams 1997); and (5) they are difficult to update when new data is obtained.

Stochastic models treat the facility deterioration process as one or more random variables that capture the uncertainty and randomness of this process. Markovian models are the most common stochastic techniques that have been used extensively in modeling the deterioration of infrastructure facilities (Butt et al. 1987; Jiang et al. 1988). These models use the Markov Decision Process (MDP) that is based on the concept of defining states of facility conditions and obtaining the probabilities of facility condition transition from one state to another during one inspection period (Jiang et al. 1988). Although Markovian models have addressed two problems in the deterministic models by capturing the uncertainty of the deterioration process and accounting for the current facility condition in predicting the future one, they still suffer from the following limitations: (1) Markovian models assume discrete transition time intervals, constant bridge population, and stationary transition probabilities, which are sometimes impractical (Collins 1972); (2) Markovian models currently implemented

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Table 1. Categories of Bridge-Deterioration Models

Category	Technique	Method
Deterministic models	Straight-line extrapolation	—
	Regression models	Stepwise regression
		Linear regression
		Nonlinear regression
	Curve-fitting models	B-spline approximation
		Constrained least squares
Stochastic models	Simulation models	—
	Markovian models	Percentage prediction
		Expected-value method
		Poisson distribution
		Negative-binomial model
		Ordered-probit model
		Random-effects model
		Latent Markov-decision process
Artificial intelligence models	Artificial neural networks	—
	Case-based reasoning	—

in advanced BMS (e.g., Pontis and BRIDGIT) use the first-order MDP that assumes state independence for simplicity (DeStefano and Grivas 1998), so this assumption means that the future facility condition depends only on the current facility condition and not on the facility condition history, which is unrealistic (Madanat et al. 1997); (3) transition probabilities assume that the condition of a facility can either stay the same or decline, so this assumption is made to avoid the difficulty of estimating transition probabilities for facilities where treatment actions are performed (Madanat and Ibrahim 1995); (4) Markovian models cannot efficiently consider the interactive effects between the deterioration mechanisms of different bridge components (Sianipar and Adams 1997); and (5) transition probabilities require updates when new data are obtained as bridges are inspected, maintained, or rehabilitated, which is a time-consuming task.

Artificial intelligence (AI) models exploit computer techniques that aim to automate intelligent behaviors. AI techniques comprise expert systems, artificial neural networks (ANN), and case-based reasoning (CBR) among others. The use of ANN in modeling bridge deterioration has been investigated by Sobanjo (1997). A multilayer ANN was utilized to relate bridge age (in years) to the condition rating of the bridge superstructure. A more detailed investigation has been made by Tokdemir et al. (2000) to predict the bridge sufficiency rating using age, traffic, geometry, and structural attributes as explanatory variables. Although ANN has automated the process of finding the polynomial that best fits a set of data points, it still shares the problems of the deterministic models.

Each BMS has a large database of bridges with inventory, inspection, and maintenance data. This database, which is updated on a regular basis, contains valuable knowledge about actual bridges that can be utilized in predicting the future condition of other bridges. This paper introduces the AI technique of CBR that makes the best use of this knowledge. CBR is expected to eliminate some of the shortcomings of the current bridge-deterioration models and provide BMS with a realistic, accurate, and generic deterioration model.

The paper is organized as follows. The first section introduces the CBR approach and explains how it can model bridge deterioration. The next section briefly presents the CBR system that was developed specifically to generate CBR applications for modeling infrastructure deterioration and the steps followed in building an application. Then, the bridge data obtained from the Canadian Province of Quebec and used in building a CBR application for modeling bridge deck deterioration is presented. The following section shows how this data was analyzed to verify its consistency and completeness and to obtain the knowledge required for building the application. Finally, the testing procedures and the validation results are discussed.

Case-based Reasoning

CBR is an AI technique, which looks for previous cases (examples) that are similar to the current problem and reuse them to solve the problem. These cases, which are stored in the so-called case library, are defined as instances that record problem definitions and their corresponding solutions. The use of CBR in modeling bridge deterioration is based on the assumption that two bridges that are similar in physical features (e.g., material, structural system, cross section, span, etc.), in environmental and operational conditions (e.g., region, highway class, traffic, load, etc.), and in inspection and maintenance history will have similar performances. This assumption will be justified in this paper through the development of a ''proof of concept'' CBR application. Although the probability of having bridges that are similar in their physical features, environmental and operating conditions, and inspection and maintenance records is not high, carrying out the matching process at the level of bridge components increases the probability of finding similar cases.

The use of CBR in modeling bridge deterioration consists of the following four steps. First, the bridge whose future conditions need to be predicted is selected as the query case, while all other bridges available in the BMS database represent stored cases in a case library. Fig. 1 shows an example of a query case that represents a bridge with an age $Aq=35$ years and inspection and maintenance records for a period $Pq=15$ years measured from now (time = 0). The goal is to predict the condition of the query case after 5 years (i.e., prediction period $Y=5$).

Second, the case library is searched to find the most similar stored case(s) to the query case—a process called case retrieval. Case indices are first used to narrow down the search space. The attributes of the query case that represent deterioration factors are then compared with those of each remaining case. In this com-

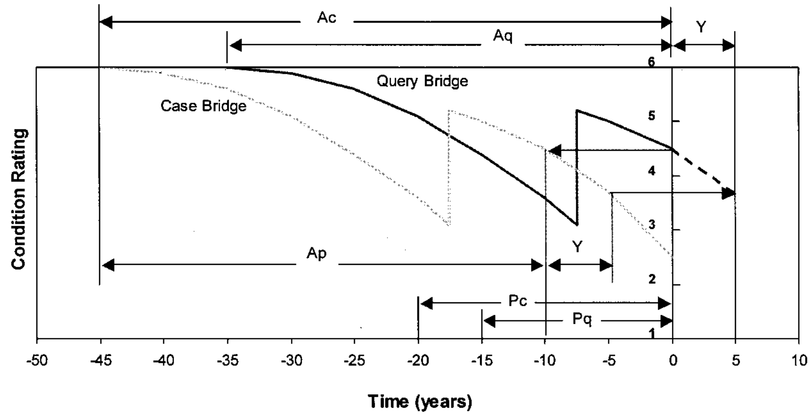

Fig. 1. Matching condition history of two bridges

parison, the degree of similarity between the values of each attribute in both the query case and a stored case is evaluated using similarity measures. A similarity measure is a function, table, matrix, or tree that defines how close the values of an attribute are to each other and that results in a value between 1 and 0 where 1 is totally similar and 0 is completely dissimilar (Morcous et al. 2002). For instance, the similarity measure of the traffic volume attribute is a linear function that ranges from 1 to 0, while the percentage of the absolute difference between the query case value and the stored case value ranges from 0 to 20%. The overall case similarity, which represents how close the stored case is to the query case, is calculated using the nearest-neighbor approach. In this approach, attribute weights, which represent the relative importance among attributes, are used to aggregate the degrees of similarity of all attributes.

Fig. 1 shows an example of a stored case with an age $Ac=45$ and inspection and maintenance records for a period $Pc=20$ measured from now. In order to retrieve this stored case as a case similar to the query case, the following requirements have to be satisfied:

1. The stored case has similarity with the query case in terms of the attributes that represent bridge deterioration factors such as structural system, region, traffic, etc.
2. There is an age called the prediction age (in the example given above, $Ap=35$) at which the condition of the stored case matches the current condition of the query case.
3. Condition records of the stored case are available for at least Y years after Ap.
4. Inspection and maintenance records of the stored case during the period from $(Ac-Pc)$ to Ap have similarity with the corresponding records of the query case during the period from $(Aq-Pq)$ to the present.

Having these requirements satisfied, the condition of the stored case at an age $(Ap+Y)$, which is 3.7 in this example, can be used as the prediction of the future condition of the query case after Y years.

Third, if the retrieved case(s) does not match the query case perfectly, the retrieved case can be modified to compensate for the differences in attribute values between the query case and the retrieved case. In this process, which is called case adaptation, domain-specific knowledge is applied to modify the retrieved case in order to increase its similarity with the query case and, consequently, provide a more accurate prediction. For example, consider that the query case is a bridge deck that is simply supported and the retrieved case is a bridge deck that is continuous at both ends. An adaptation rule of the form IF-THEN can be used to reduce the condition of the retrieved case and compensate for the difference in the structural continuity between the query case and the retrieved case.

Fourth, when new data about existing bridges (e.g., inspection and maintenance data) becomes available or new bridges are built, the case library is augmented or updated with the new data. This process is called case accumulation, which extends the coverage of the case library and increases the probability of finding similar cases. Although CBR may appear similar to database management systems (DBMS), it has some important differences (Leake 1996):

1. CBR supports the representation and indexing of cases that are complex in structure and composed of many subcases. Relevant subcases are retrieved individually from the case library and aggregated to provide a problem solution. Relational DBMSs, which are the most popular, have relatively flat structures in comparison.
2. Unlike DBMSs, which support exact matching only, CBR supports partial matching and assesses the similarity among cases.
3. CBR is able to incorporate domain-specific knowledge that supports the adaptation of retrieved cases.
4. CBR supports comparing dynamic data (i.e., time-dependent data) as well as static data.

The CBR approach provides BMSs with some advantageous characteristics:

1. It can consider the condition history of bridge components depending on the availability of condition data. This minimizes the effects of the uncertainty and randomness of the deterioration process because of its ability to capture the effects of unobserved deterioration factors.
2. It can perform "what if" analysis for different maintenance scenarios by changing maintenance decisions and retrieving cases with similar decisions when maintenance data are recorded.
3. It uses specific, nongeneralized knowledge encapsulated in previous cases to predict the future condition of other cases which is more accurate (Leake 1996).
4. The model is updated as a natural by-product of case and data accumulation in the BMS database.
5. It can recognize the interaction among different bridge components by considering the condition of these interacting components as a part of the case description.

6. It can manipulate cases with variable inspection periods and aggregated (i.e., continuous) condition ratings.

However, it should be mentioned that the success of the CBR approach depends on the size of the case library, the adequacy of case description, the correct setting of the attribute weights and degrees of similarity, and the availability of adaptation knowledge, which may limit its applicability in certain domains. The CBR application presented later will demonstrate these characteristics except the ''what if'' analysis because of the unavailability of maintenance data.

Building Case-based Reasoning Application

Case-based reasoning for modeling infrastructure deterioration (CBRMID) is a new CBR system that was developed specifically to generate CBR applications for predicting the future conditions of facilities in any infrastructure domain. Case-based reasoning for modeling infrastructure deterioration has distinct characteristics that satisfy the special requirements for modeling infrastructure deterioration such as representing infrastructure facilities with deep hierarchical decomposition; considering the interaction among facility components in the matching process; allowing the versatility and extensibility of both case structure and contents; supporting data reusing and sharing; representing facility time-dependent data; and accounting for the fuzziness of retrieval knowledge. For more information about CBRMID, please refer to Morcous et al. (2002). Below are the four steps for developing a CBR application using CBRMID. These steps are followed in developing the ''proof of concept'' CBR application presented in the next section.

1. Construct the domain model: A domain model is the conceptual model of the domain under consideration (e.g., bridges, culverts, buildings, etc.). It consists of real-world concepts (physical or nonphysical) that use the vocabulary of the domain and are meant for domain personnel not for computer specialists (Rivard and Fenves 2000). Relationships among the domain concepts are also identified in the domain model. The main function of such a model is to define how cases are represented and accumulated in the case library. Since the domain model is not static, it can be expanded with additional concepts and new relationships even after the relevant cases are created.
2. Define attributes: Attributes are the features used to describe the cases and their conditions. Attributes are defined for each concept in the domain model constructed in Step 1. Properties of these attributes, such as attribute type, weight, role, possible values or ranges, and similarity measures have to be defined. These properties represent the knowledge required by the system for case retrieval.
3. Define adaptation rules: Adaptation rules are IF-THEN expressions that encapsulate domain-specific knowledge obtained from domain experts and literature to be used in case adaptation. Adaptation rules are defined for each domain concept as a set of rule conditions and actions using the attributes defined in Step 2.
4. Construct the case library: Cases are the primary source of knowledge in CBR systems. Cases may be complex in their structure and composed of many components. Constructing the case library consists of defining the structure of each case (e.g., each bridge) and storing data about its components. These data, which may be static or dynamic in nature, are represented as values for the attributes defined in Step 2.

Case-based Reasoning Application for Modeling Deck Deterioration

The ''proof of concept'' CBR application presented in this paper focuses on modeling the deterioration of concrete bridge decks because the concrete bridge deck is the bridge component that has the highest deterioration rate due to its direct and continuous exposures to traffic, weather, and deicing chemicals (Freyermuth et al. 1970). Moreover, data about concrete bridge decks are more readily available than any other bridge component in the literature on bridge deterioration. Modeling the deterioration of concrete bridge decks is sufficient to prove the concept since the same procedures can be applied to other bridge components.

The case library of the developed CBR application was stocked with cases obtained from the database of the Ministry of Transportation of Quebec (MTQ). This database is part of a comprehensive system for managing various highway structures. It accounts for 9,500 province-owned highway structures that are grouped into eight categories: culverts, slab bridges, beam bridges, box-girder bridges, truss bridges, arch bridges, cabled bridges, and other structures. This database includes three types of data for each highway structure:

1. Inventory data—which consists of approximately 220 attributes that can be categorized into (MTQ 1997) administrative data (e.g., identification, location, jurisdiction, etc.), technical data (e.g., environment, traffic, postings, etc.), and descriptive data (e.g., geometry, material, structural system, etc.).
2. Inspection data—which consists of detailed visual inspections collected for each structure every 3 years (or less) since 1993. These data are collected using 21 different inspection forms, each of which corresponds to a group of correlated structural elements such as foundation elements, truss elements, and deck elements. Each inspection form includes the material condition rating (MCR), which represents the condition of an element based upon the severity and extent of observed defects, and the performance condition rating (PCR), which describes the condition of an element based upon its ability to perform its intended function in the structure (MTQ 1995). Both MCR and PCR are represented in an ordinal rating scale that ranges from 1 to 6, where 6 represents a new condition. Table 2 shows how the MCR of an element in a highway structure in Quebec is determined given the type of element (i.e., primary, secondary, or auxiliary) and the percentage of severe material defects in the element cross section, surface area, or length.
3. Maintenance data—which represents the major maintenance actions that are recommended to take place in the future. The cost and the expected time of each action are roughly estimated. Data about maintenance actions that took place in the past are collected by the regional offices, each with its own practice, and not recorded in the central MTQ database. Thus, even though CBRMID is able to consider the effect of previous maintenance actions on the predicted conditions, these data were not entered in the case library because of the difficulty of having access to it.

As a first step in building a CBR application for modeling bridge deck deterioration, a domain model that consists of five domain concepts was defined. This domain model is a subset of a more comprehensive one that represents all the concepts of the bridge-management domain (Morcous 2000). The five concepts of the domain model, listed in Table 3, are bridge, bridge inspection, deck, girder, and element condition. These concepts are connected to each other by different types of relationships such as

Table 2. Material Condition Rating System used in Quebec (MTQ 1995)

MCR	Percentage of Very Severe Material Defects in Element Cross Section, Surface Area, or Length: Primary element	Secondary element	Auxiliary element
6	<1.5	<3	<6
5	1.5–5	3–10	6–20
4	5–10	10–20	20–40
3	10–15	20–30	40–60
2	15–20	30–40	60–80
1	>20	>40	>80

containment relationships (e.g., between bridge and deck) and association relationships (e.g., between bridge and bridge inspection).

As a second step in building the CBR application, the attributes of the concepts mentioned were defined based on a previous study carried out by Carrier and Cady (1973) on bridge deck deterioration and a literature review carried out by the writers on the deterioration factors recognized by state-of-the-art BMS. These attributes are listed in Table 3. The four attributes, total width, number of beams, construction date, and inspection date, do not participate in the retrieval process directly because they are used in deriving the values of the two attributes, beam spacing and age, that do participate in the retrieval process. The MCR attribute is the only stored attribute that is time-dependent. This attribute is used twice: (1) by the deck concept to describe the condition history of bridge decks and predict their future conditions; and (2) by the girder concept to describe the condition history of bridge girders and account for the interactive effects between the deterioration mechanisms of a deck and its supporting girders. Table 3 also lists the possible attribute values as ranges for numeric attributes and as enumerated values for symbolic attributes. It should be noted that the values of the region attribute (i.e., northern, central, eastern, and western regions of Quebec) are categorized based on the geographical locations only and regardless of the climatic characteristics of each region.

The third step in building a CBR application is case adaptation. Although CBRMID supports case adaptation by substitution (Morcous et al. 2002), this step was skipped in building the application. This is because the main focus of this paper is to prove the ability of CBR to retrieve relevant cases and reuse their condition data in prediction. Case adaptation will be considered in future research since it may have positive impacts on the prediction.

Concrete bridge decks in beam bridges were selected to construct the case library of the application because this type of bridge is dominant in Quebec and it provides the majority of concrete decks. The MTQ database was screened by eliminating bridge records with incomplete inventory data or missing condition data to avoid misleading validation results. Furthermore, bridge records that have positive deterioration rates were also eliminated because they indicate that major maintenance actions have been made during the observation period and the corresponding maintenance data are not available in the database. These records can be incorporated into the case library later when their maintenance data become available. This screening resulted in 240 bridges with a total of 521 bridge decks that were entered in the case library as the fourth and last step in building the proposed application.

Data Analysis

A statistical data analysis was carried out in order to evaluate the quality (i.e., the consistency and completeness) of the data set selected for building the case library (i.e., the 521 bridge decks) and to obtain retrieval knowledge (i.e., relative importance of attributes and degrees of similarity among attribute values). Two different statistical techniques were used to assess the association between the variables according to their types: (1) measuring the coefficient of correlation for quantitative variables, and (2) performing the analysis of variance (ANOVA) test for qualitative variables. Only for the purpose of this statistical analysis, the rate of deterioration between successive deck inspections was selected to be the response variable because the value of this variable is slightly affected by the maintenance actions taken in the past. The bridge deck condition was not selected because the value of this variable is greatly affected by the maintenance actions taken in the past, which are not readily available. Moreover, incremental

Table 3. Attributes used in Describing Bridge Deck Cases

Domain concept	Attribute	Actual value/range	Derivation
Bridge	Region	Northern, central, eastern, western	Stored
	Highway class	Express, national, regional, collector, local	Stored
	Average daily traffic (ADT) (pcu)	0–136,000	Stored
	Truck percent (ADTT/ADT)	0–38	Stored
Bridge inspection	Inspection date (year)	1993–1999	Stored
Deck	Span length (m)	5.8–39.70	Stored
	Skew angle (degree)	0–54.94	Stored
	Material	Concrete	Stored
	Total width (m)	4.7–33.36	Stored
	Beams spacing (m)	1.07–5.55	Derived
	Structural system	Simple, continuous	Stored
	Wearing surface	Cement mix, bituminous mix	Stored
	Construction date (year)	1923–1996	Stored
	Number of beams (unit)	2–16	Stored
Girder	Material	RC, PC, (precast)	Stored
Element condition	Material condition (rating)	1–6	Stored
	Age (year)	1–76	Derived

Table 4. Correlation Matrix of Eight Quantitative Deterioration Factors

Factor	Average daily traffic	Truck percent	Span length	Skew angle	Beam spacing	Age	Deck material condition rating	Girder material condition rating
Truck percent	−0.11							
	0.00							
Span length	0.21	−0.01						
	0.00	0.76						
Skew angle	−0.06	0.21	−0.01					
	0.07	0.00	0.76					
Beam spacing	0.14	−0.05	−0.01	−0.06				
	0.00	0.16	0.75	0.08				
Age	−0.25	−0.16	−0.45	−0.07	−0.02			
	0.00	0.00	0.00	0.06	0.62			
Deck material condition rating	0.12	0.03	0.34	−0.18	−0.02	−0.39		
	0.00	0.40	0.00	0.00	0.65	0.00		
Girder material condition rating	0.08	0.10	0.16	−0.06	−0.01	−0.40	0.54	
	0.03	0.01	0.00	0.08	0.77	0.00	0.00	
Deterioration rate	−0.17	0.04	−0.09	0.02	0.04	0.11	0.02	−0.03
	0.00	0.25	0.02	0.50	0.23	0.00	0.55	0.37

variables that represent changes in condition over time are more realistic than those that represent the condition itself (Madanat and Ibrahim 1995). In order to have positive values of the response variable, which lead to easier interpretation of the results, the rate of deterioration is calculated as follows:

$$R_i = \frac{\mathrm{MCR}_i - \mathrm{MCR}_{i+1}}{t_{i+1} - t_i}$$

where R_i = rate of deterioration between inspection i and inspection $i+1$; MCR_i = material condition rating at inspection i; and t_i = time at inspection i.

The MCR used in the previous equation is a real-valued variable (i.e., cardinal attribute) that represents the overall deck condition. The value of this variable is calculated as the aggregation of MCR of the different deck parts: two exterior faces, two end portions (each portion has a length equal to two-times the girder depth), and a middle portion. These MCRs are aggregated using the so-called balance factors listed in MTQ (1995). These factors represent the relative importance among the different parts of a single component. The calculated rate of deterioration is also a real-valued variable that represents the decline in the overall deck condition in 1 year. This variable differs slightly from the one reported by Madanat and Ibrahim (1995) where they used a discrete incremental variable that represents the number of drops in the deck condition states during one inspection period.

The total number of records used in the data analysis was 780 due to the fact that every two records of deck inspection result in one record of rate of deterioration. It should be mentioned that the inspection period in this data set is not constant but it has an average of 2.8 years and a standard deviation of a 0.73 year.

Table 4 shows the correlation matrix (only the lower triangle is shown because of symmetry) that displays the coefficient of correlation (shown in the top of each cell) between each pair of quantitative attributes and the p-value (shown below the coefficients of correlation) for the test of hypothesis that the coefficient of correlation is equal to zero. Using a level of significance, $\alpha = 0.05$, all coefficients of correlation that have p-value greater than or equal to 0.025 are considered insignificant. The last row in the correlation matrix contains the associations of all quantitative variables with the response variable "deterioration rate." Only the associations with the "average daily traffic" (ADT), "span length," and "age" are significant. The negative association between the "deterioration rate" and the "ADT" indicates more stringent design requirements and adequate preventive maintenance of the highway structure that have higher traffic volumes. The negative association between the "deterioration rate" and the "span length" indicates that bridges with shorter spans have a higher deterioration rate than bridges with longer spans. This can be explained by knowing that 90% of the bridge spans whose length is less than 20.0 m are made of reinforced concrete girders, which cause decks to deteriorate faster, while 75% of the bridge spans whose length is greater than 30.0 m are made of prestressed concrete girders, which cause decks to deteriorate slower. The positive association between the "deterioration rate" and the "age" indicates that older bridge decks have higher deterioration rates, which is in agreement with previous research (Bulusu and Sinha 1997). Although the coefficient of correlation between the "deck MCR" and the "girder MCR" has the highest value in the matrix, this should not be considered as a result of component interaction only since the two components are affected by the same underlying factors (i.e., multicollinearity).

Table 5 shows the results of five one-way ANOVA tests that were carried out to measure the association between the response variable and five qualitative variables: "highway class," "region," "material," "structural system," and "wearing surface." In each test, some descriptive statistics about the deterioratione rate for each value of the qualitative explanatory variable are calculated along with the significance of the test represented by the p-value of the F test that compares the variation due to the qualitative variable with the variation due to errors. This test shows that the "highway class" and "region" are the only significant variables among the five explanatory variables (using a level of significance $\alpha = 0.05$). The low deterioration rate of bridges on express highways confirms the fact that these bridges have more stringent design requirements and a higher priority for preventive maintenance than bridges on other highway classes because of their importance to public use. Table 5 shows that bridge decks in western Quebec have the lowest deterioration

Table 5. Analysis of Variance Tests of Five Qualitative Deterioration Factors

Attribute	Values	N	Mean	Standard deviation	F	P
Highway class	Collector	82	0.102	0.124	7.66	0
	Express	145	0.032	0.060		
	Local	167	0.095	0.131		
	National	258	0.087	0.134		
	Regional	128	0.109	0.179		
Region	Central	227	0.099	0.138	10.9	0
	Eastern	255	0.106	0.143		
	Northern	168	0.070	0.122		
	Western	130	0.032	0.096		
Material	Cast-in-place prestressed	55	0.064	0.124	2.44	0.088
	Cast-in-place reinforced	555	0.090	0.140		
	Precast prestressed	170	0.069	0.111		
Structural system	Continuous	216	0.090	0.154	0.68	0.408
	Simple	564	0.081	0.124		
Wearing surface	Bituminous mix	742	0.085	0.135	2.15	0.143
	Cement mix	38	0.053	0.082		

rate, while those in eastern Quebec have the highest deterioration rate.

Case-based Reasoning Validation

Three different methods for validation were found in the literature on CBR applications. In Roddis and Bocox (1997), a CBR application for detecting steel-bridge fabrication errors (CB-BFX) was validated by comparing its solution with the solutions provided by a knowledge-based expert system that has been in use for many years. In Ng and Smith (1998), a CBR application for evaluating contractor prequalification (EQUAL) was validated by comparing its solutions with the solutions provided by domain experts and semiexperts (i.e., a Turing test). In Arditi and Tokdemir (1999), a CBR application for predicting the outcome of construction litigation was validated by comparing its solutions with the solutions of real-world cases that were not stored in the case library. The last method of validation was adopted here because it makes its comparisons with actual cases and not with human judgement, which is subjective, or with other systems, which may be erroneous. This section describes the procedures followed in validating the accuracy of the solutions provided by the developed CBR application.

Out of the 521 bridge decks used in the data analysis, a subset of 259 bridge decks (i.e., cases) was used in the validation process because they had three inspection records, which is the minimum number of records required to test the matching of the condition history. Two groups of cases, each consisting of 30 cases, were selected at random from this subset: the first group is used to refine the application parameters (i.e., training group), while the second group is used to validate the application (i.e., testing group). The last inspection records containing the latest bridge deck conditions in each of these two groups were removed and used to validate the accuracy of the solution provided by the CBR application. Condition records of the other decks that belong to the same bridges of the training and testing cases are also removed to avoid retrieving decks from the same bridges. For each group, two different sets of attribute weights, which represent the relative importance among different attributes, were used. In the first set, the same numeric value was assigned to all attribute weights assuming that all attributes have the same importance. The reason for using equal attribute weights is to study the effect of adjusting attribute weights on the validation results. In the second set, attribute weights were adjusted manually according to their relative importance based on the knowledge obtained mainly from the data analysis carried out in the previous section. For example, the weight of the ‘‘region’’ attribute was set to 0.8 (i.e., very important), the weight of ‘‘girder MCR’’ was set to 0.6 (i.e., important), and the weight of the ‘‘truck percent’’ attribute was set to 0.2 (i.e., slightly important). Attribute weights were refined in several iterations until the results obtained with the training group were satisfactory. Degrees of similarity among attribute values were also estimated based on the knowledge obtained mainly from the data analysis and used with both equal and adjusted attribute weight sets.

Three different methods for calculating the predicted value from the results of retrieved cases were examined to identify the most successful calculation method. Each of these methods makes use of the overall case similarity of each retrieved case. These methods, as reported by Arditi and Tokdemir (1999), are as follows:

- Method 1: the true condition is compared to the condition of the retrieved case that has the highest overall case similarity.
- Method 2: the true condition is compared to the most frequent condition found in the top ten retrieved cases (or fewer if ten are not available) that have an overall case similarity greater than or equal to 0.75. The condition rating scale was divided into ten intervals (interval width equal to 0.25) and the interval that contains the highest number of retrieved cases is the one considered.
- Method 3: the true condition is compared to the average condition of the top five retrieved cases (or fewer if five are not available) that have an overall case similarity greater than or equal to 0.75. The average of the predicted conditions is weighted using the overall case similarities to magnify the importance of the retrieved cases that have higher similarities.

Table 6 shows the results of testing the developed application with the two groups of cases (training and testing groups) and using both equal and adjusted attribute weights. These results were obtained by allowing a tolerance value of 0.25, which is the maximum value permitted for the absolute difference between the actual condition (i.e., observed) and the predicted condition. The prediction is considered ‘‘correct’’ if the predicted condition is

Table 6. Testing Results for Equal and Adjusted Attribute Weights

Category	EQUAL WEIGHTS				ADJUSTED WEIGHTS						
Data Group	Training		Testing		Training		Testing		Average		
Calculation method	Percent retrieved	Percent correct	Percent retrieved	Percent correct	Percent retrieved	Percent correct	Percent retrieved	Percent correct	Percent retrieved	Percent correct	Percent retrieved correct
1	100	70	100	83	100	77	100	90	100	80	80
2	60	61	70	86	77	78	83	80	73	76	55
3	60	83	70	86	77	78	83	84	73	83	60
Average	73	71	80	85	84	78	89	85	82	80	65
Percent retrieved correct	52		68		66		76		n/a		

within the 0.25 tolerance of the observed condition. The tolerance value is determined based on the user needs and the rating system used.

In Table 6, the ''percent retrieved'' column gives the percentages of query cases for which similar cases were retrieved, whether with correct or incorrect predictions. The ''percent correct'' column gives the percentages of those retrieved cases that provided correct predictions. These percentages are listed for the training and testing groups using each of the three calculation methods presented earlier and for both equal and adjusted weight sets. For example, in the testing group with equal weights, Calculation Method 3 retrieved cases for 21 query cases out of 30 cases (''percent retrieved'' = 70) and predicted accurately 18 query cases out of these 21 cases (''percent correct'' = 86). The ''percent retrieved correct'' value represents the percentage of the query cases that obtained retrieved cases with correct predictions, which is the multiplication of ''percent retrieved'' and ''percent correct'' values. The average of the ''percent retrieved,'' the average of the ''percent correct,'' and the average of the ''percent retrieved correct'' are calculated for the three calculation methods on the right-most columns and for each testing group with both equal and adjusted attribute weights on the lowest row.

The effect of adjusting attribute weights is manifested in the increase of the average percentage of retrieved correct solutions from 52 to 66 for the training group, and confirmed by the increase of the average percentage of correct solutions from 68 to 76 for the testing group (shown in the lowest row). It is clear that adjusting attribute weights is important and provides better results than equal weights. Establishing proper attribute weights is not a simple task since it requires performing statistical analysis and/or eliciting knowledge from domain experts. This task is carried out once while building the CBR application for the first time and may require some updates if drastic changes in the bridge network have occurred.

The average of the percentages of retrieved and correct solutions varies according to the calculation method used. This variation is shown in the ''average'' column at the right of Table 6. Although Method 3 fails sometimes to give any solution (''percent retrieved'' = 73), it provides the highest percentage of correct solutions out of the retrieved cases (''percent correct'' = 83). Method 1 has the highest percentage of retrieved cases (''percent retrieved'' = 100) because it retrieves the solution of the best-matched case regardless of its degree of similarity, but it has a relatively lower percentage of correct solutions (''percent correct'' = 80) than Method 3. Method 2 has the same percentage of retrieved cases as Method 3 because it has the same minimum acceptable degree of similarity (i.e., 0.75), but it has a lower percentage of correct solutions (''percent correct'' = 76). From these results, it can be concluded that Method 3 is preferred when cases can be retrieved with a minimum acceptable degree of similarity otherwise Method 1 can be used

The overall accuracy of the developed CBR model with Method 3 was estimated at 84% (i.e., in the ''percent correct'' column for the testing group with adjusted weights). Since Method 3 found similar cases for only 83% of the cases (''percent retrieved''), the accuracy for the whole testing group was computed at 70% by multiplying 84 by 83%. An interval estimate for this accuracy was calculated using a confidence interval equal to 95% and a sample size equal to 30. This estimate showed that the accuracy of the CBR model ranges from 54 to 86%.

To evaluate the performance of the CBR model with respect to the performance of deterioration models employed by current BMSs, a multiple nonlinear regression model was developed using the explanatory variables presented earlier in the data analysis as entry variables and the rate of deterioration as the response variable (i.e., incremental model). This incremental model predicts the future condition based on the estimated deterioration rate and the most recent condition observation. Although other types of regression models may have better prediction capabilities, this type of common deterioration model was used because of its simplicity and widespread use. Using the same subset of data utilized in building the CBR case library, the explanatory variables were screened by using the stepwise regression technique and choosing a minimum value of 3.0 for the F-statistic of their test of hypothesis. The results of testing the CBR model and the commonly-used regression model are shown in Fig. 2 for different tolerance values. The tolerance value represents the maximum acceptable value for the absolute difference between the predicted condition and the actual condition—the lower the tolerance value, the higher the prediction accuracy. Fig. 2 shows clearly that the CBR model provides more accurate predictions even when low tolerance values are allowed (e.g., 0.25). This accuracy may assist decision makers in the respective agencies to improve the reliability of their MRR decision and, consequently, optimize the efficiency of their management systems.

Conclusions

Modeling bridge deterioration is an essential task in making MRR decisions for a network of bridges. However, bridge-deterioration models currently employed in BMSs have some limitations that

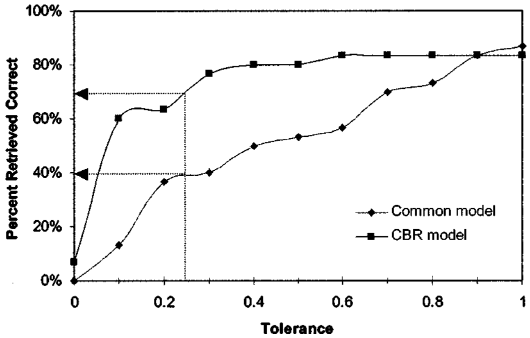

Fig. 2. Comparing accuracy of case-based reasoning and commonly used regression models

affect the quality of these decisions such as disregarding the effects of bridge-condition history on the predicted condition and overlooking the interactive effects among different deterioration mechanisms of bridge components. Moreover, updating these models when new data are obtained is not a simple task. This paper proposed a new CBR approach to address these limitations and provide BMSs with a realistic, accurate, and generic bridge-deterioration models.

The CBR approach was validated through the development of a ''proof of concept'' CBR application that models the deterioration of concrete bridge decks. This application was developed using the CBR system called CBRMID that was developed specifically to satisfy the requirements of modeling facility deterioration in any infrastructure domain. Data about a set of 259 concrete bridge decks from Quebec provided the opportunity to construct the case library. This validation proved that the use of the CBR approach in modeling bridge deterioration has the following advantageous characteristics: (1) CBR minimizes the effect of the uncertainty and randomness in the deterioration process since it retrieved not only bridge decks that have the same physical features as the query one but also those that have similar condition history, which captures the effect of the unobserved deterioration factors; (2) CBR is not restricted to a constant inspection period or a discrete condition rating system since it retrieved bridge decks with different inspection periods and aggregated conditions; (3) CBR allows the interaction among the deterioration mechanisms of different bridge components to be considered in modeling their deterioration since it included matching the condition of the supporting girders while retrieving similar decks; and (4) CBR provides BMSs with relatively accurate predictions at the network level since the CBR model provided correct prediction in 70% of cases.

In addition to these advantageous characteristics, the CBR approach has the potential to model the deterioration of different bridge components as well as other infrastructure facilities, to performing ''what if'' analysis for different maintenance decisions, and to learn and improve its predictive power as new cases, inspection records, and maintenance actions are accumulated in the BMS database. These characteristics will be tested in future research.

In spite of the distinct characteristics of the CBR approach, it has some drawbacks that can be summarized as follows: (1) CBR may not be able to retrieve any cases when the size and coverage of the case library are inadequate; (2) the determination of attribute weights and degrees of similarity requires engineering judgement, which suffers from subjectivity; and (3) the acquisition of domain-specific knowledge for case adaptation is not a simple task.

The performance of the developed CBR model could be further improved if important deterioration factors, such as concrete cover, protective system, construction practice, use of deicing chemicals, and maintenance actions were included in the case description. Also, more accurate predictions could be obtained if adaptation knowledge was incorporated and attribute weights were automatically established. Therefore, further research is required in order to consider these enhancements, verify the compatibility of the CBR model with the other BMS modules, and validate its performance in a real-world environment. Current research efforts are being devoted to applying the CBR approach in another infrastructure domain, the domain of flat building roofs. The objective is to confirm the predictive power of the CBR approach and to compare its performance against the popular Markovian models.

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